

Physics of Semiconductor Devices

ECE210: PSD



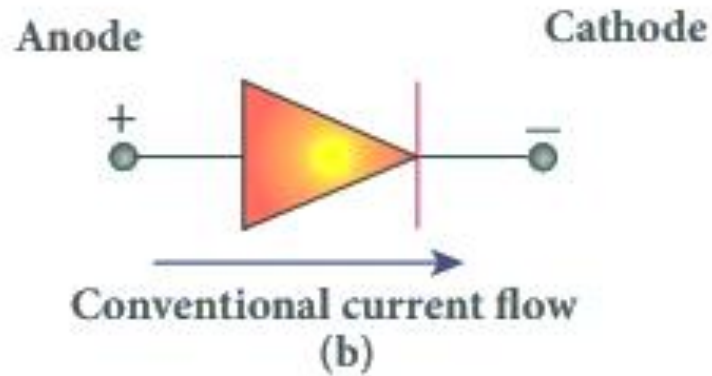
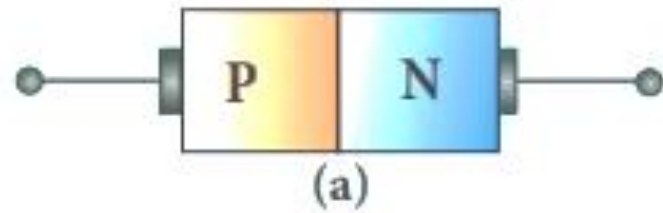
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Dr. Ram Krishna Ghosh

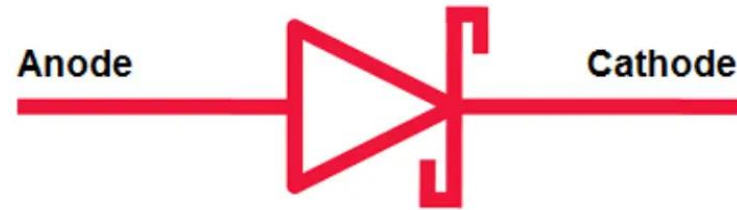
Week 04



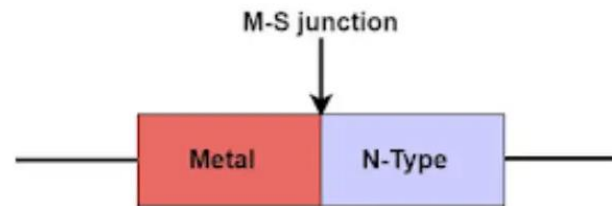
Introduction



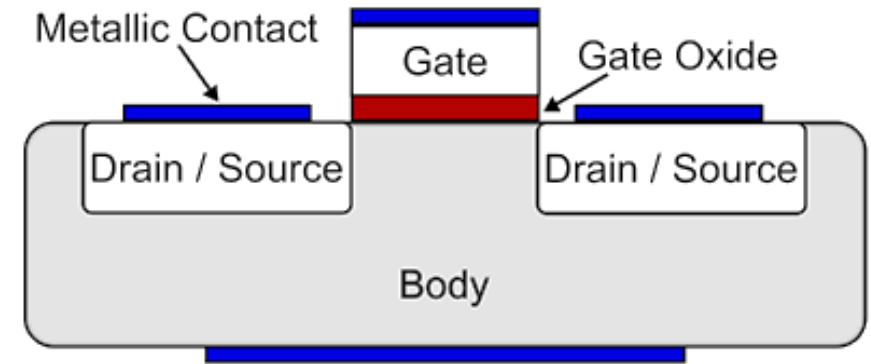
pn diode



Schottky diode symbol



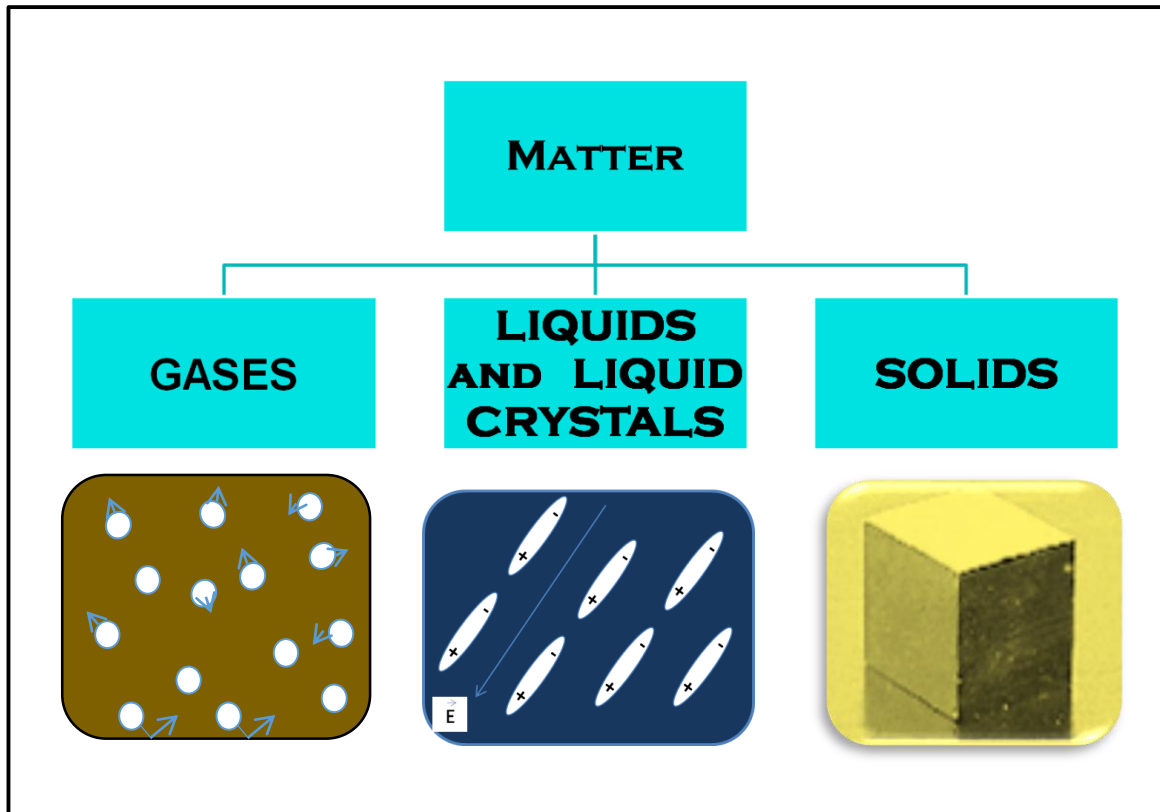
Schottky diode



MOS Transistor

Electronics industry relies on tuning of a special class of material, the semiconductor, which lies right at the metal-insulator boundary.

Introduction to crystal physics



- Matter exists in three states viz. solids, liquids and gases.
- All these states are composed of atoms and molecules.
- When we focus the solids, they are classified into many types based on several properties like electrical, mechanical, magnetic, optical, thermal etc.

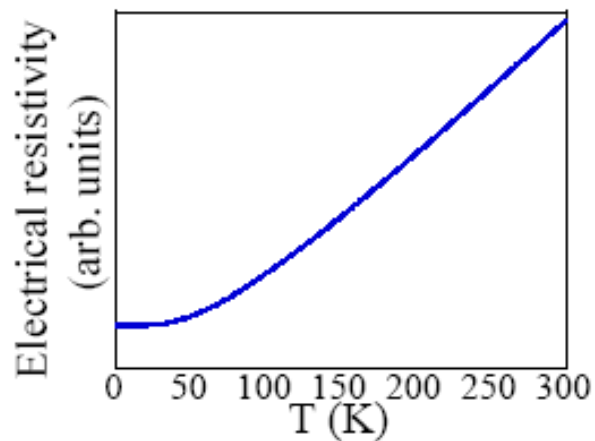
- ❑ **Solid state physics (SSP) explains the properties of solid materials as found on earth.**
- ❑ **The properties are expected to follow from Schrödinger's eqn. for a collection of atomic nuclei and electrons interacting with electrostatic forces.**

Introduction to crystal physics

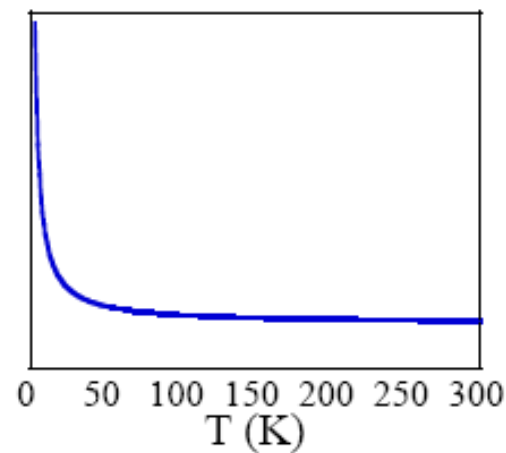


- **Graphite** is a metal, **Diamond** is an insulator and **Buckminsterfullerene** is a superconductor. How can this be? After all, they each contain a system of atoms and especially electrons of similar density.

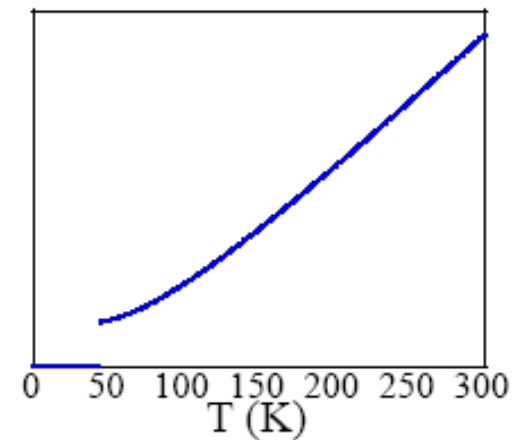
They are all just carbon!



Metal

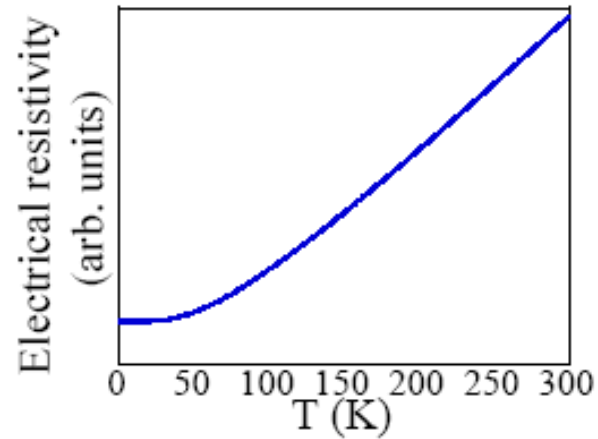


Insulator

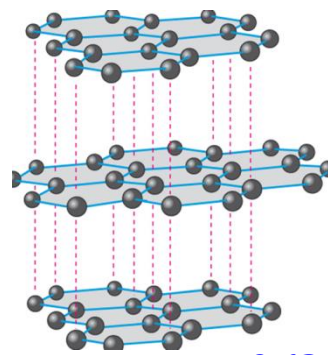


Superconductor

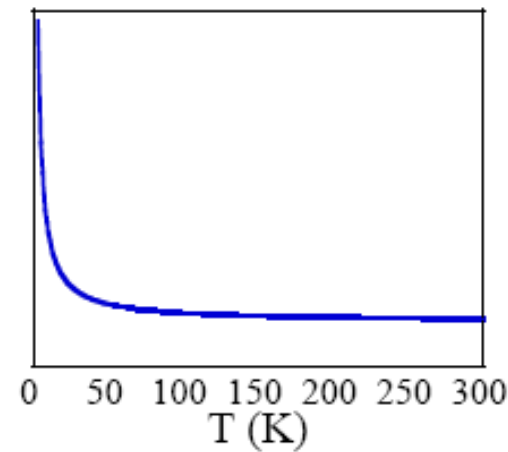
Resistance vs. Temperature



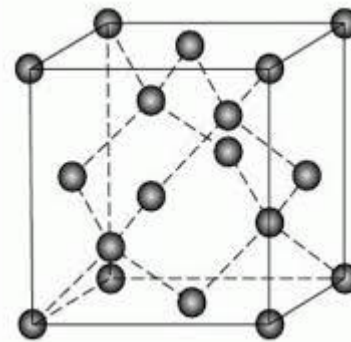
Metal



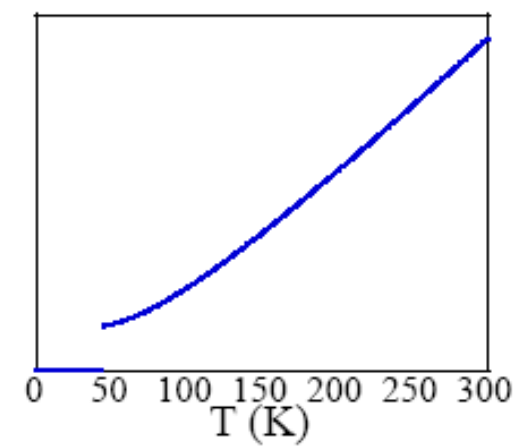
graphite



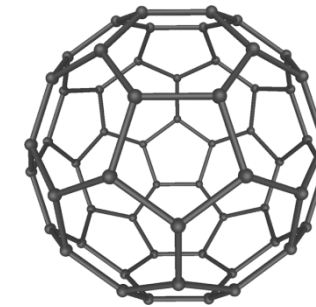
Insulator



diamond



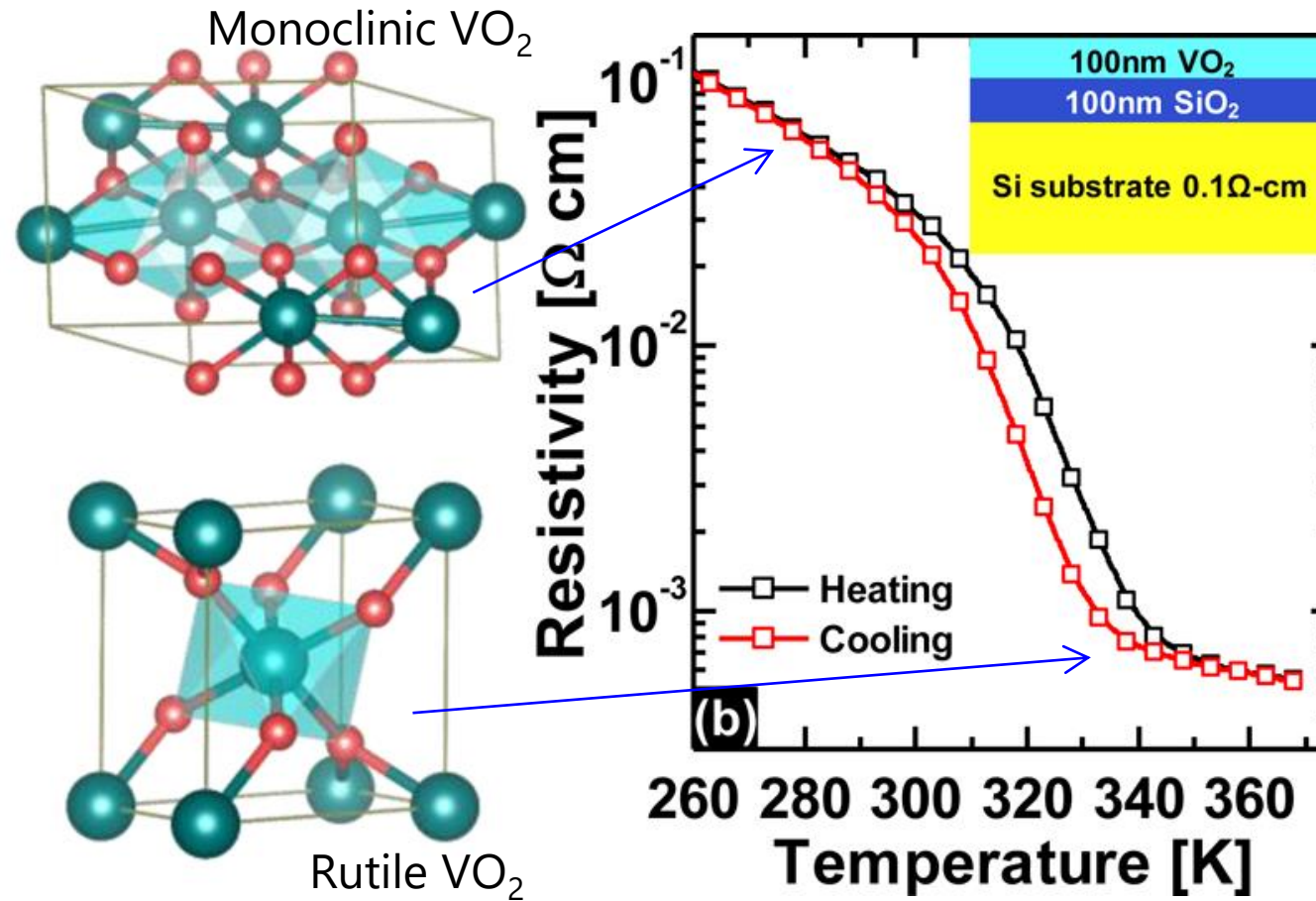
Superconductor



Alkali doped Buckminsterfullerene

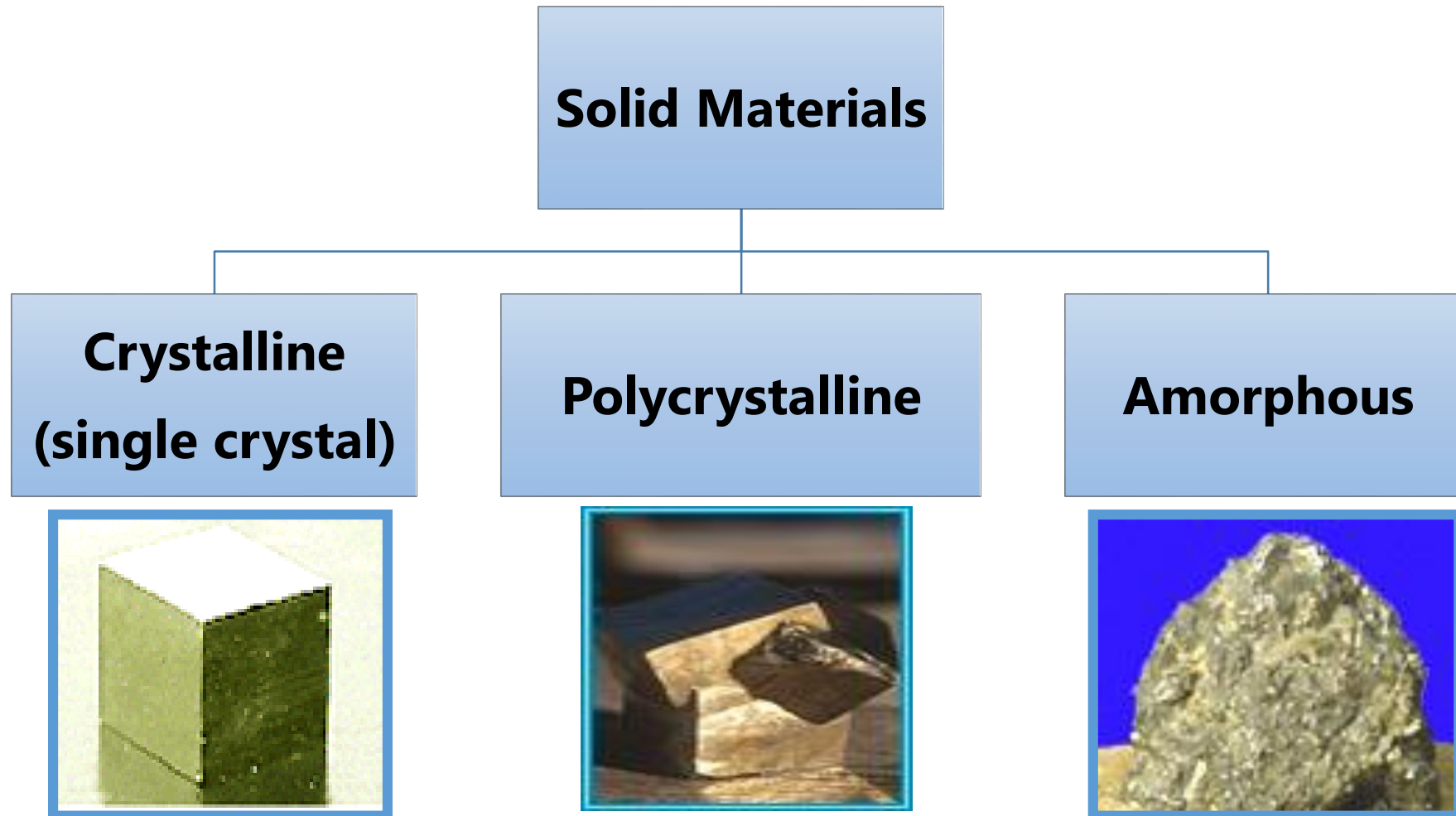
The main reason for these different properties of solids is their crystal structure

Metal-Insulator phase transition



Ref: ACS Nano, **2015**, 9 (2), pp 2009–2017

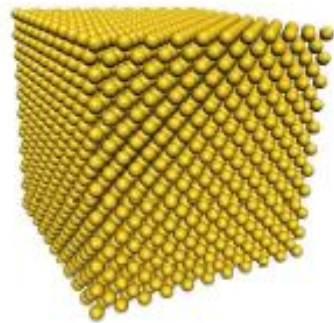
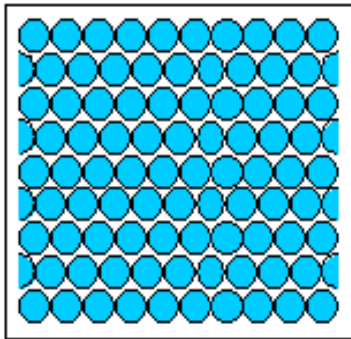
Classification of solids



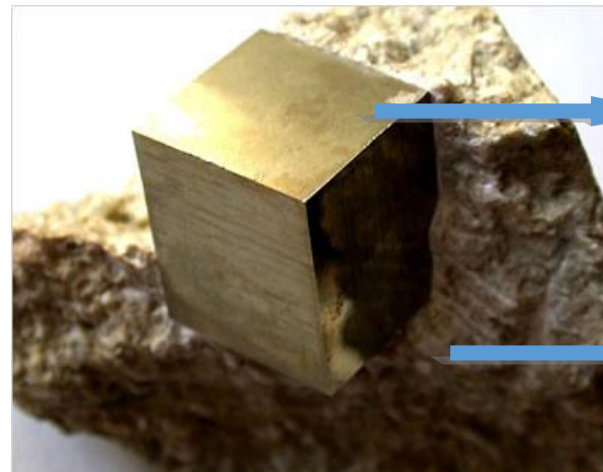
Single Crystals



- **Single crystals** have a periodic atomic structure across its whole volume.
- At long range length scales, each atom is related to every other equivalent atom in the structure by translational or rotational symmetry



Single Crystals



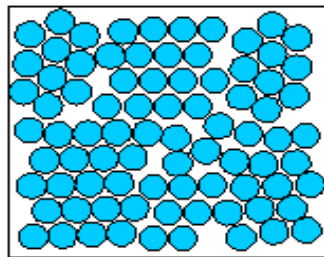
Single Pyrite Crystal

Amorphous Solid

Polycrystalline materials



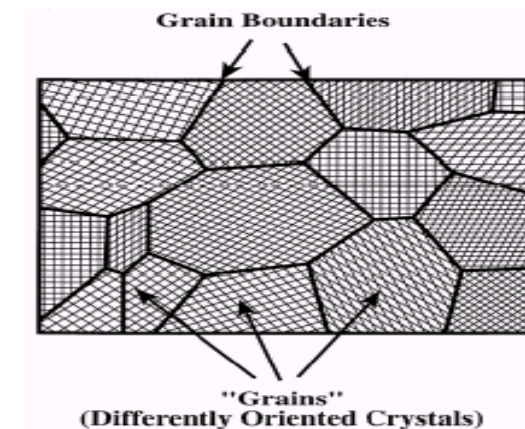
- An aggregate of many small single crystals. These regions are called as **grains (domain)** and are separated from one another by grain boundaries. The atomic order can vary from one domain to the next.
- These materials have a high degree of order over many atomic or molecular dimensions.
- These ordered regions, or single crystal regions, vary in size and orientation wrt one another.
- The **grains are usually 100 nm - 100 μ m** in diameter. Polycrystals with grains that are < 10 nm in diameter are called **nanopolycrystalline**.



Crystal Structure



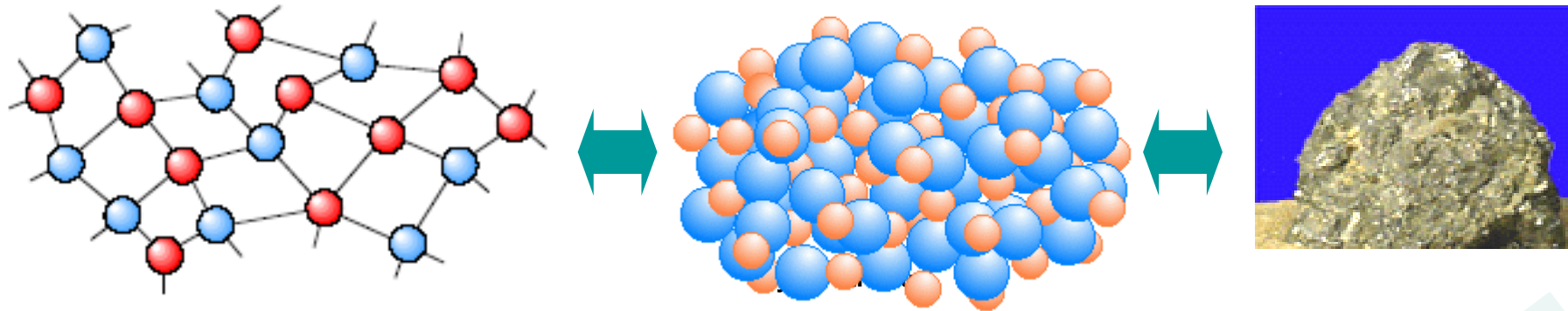
Polycrystalline
(Grain)



Amorphous materials



- Composed of **randomly orientated atoms, ions, or molecules** that do not form defined patterns or lattice structures.
- Amorphous materials have order only within a few atomic or molecular dimensions. Amorphous materials **do not have any long-range order**, but they have varying degrees of short-range order.
- Examples to amorphous materials include amorphous silicon, amorphous HfO_2 , SiO_2 , plastics, and glasses.
- Amorphous silicon can be used in solar cells and thin film transistors.



Crystal lattice

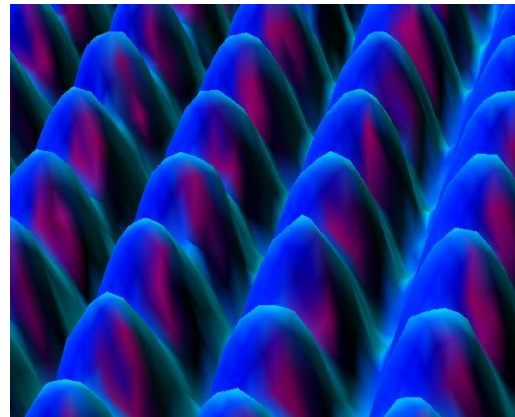


What is a crystal lattice?

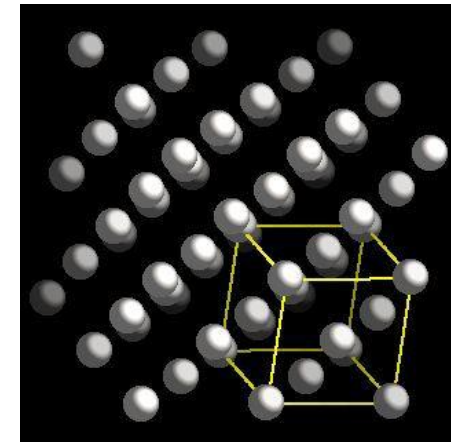
In crystallography, only the geometrical properties of the crystal are of interest, therefore one replaces each atom by a geometrical point located at the equilibrium position of that atom.



Platinum



Platinum surface
(scanning tunneling microscope)

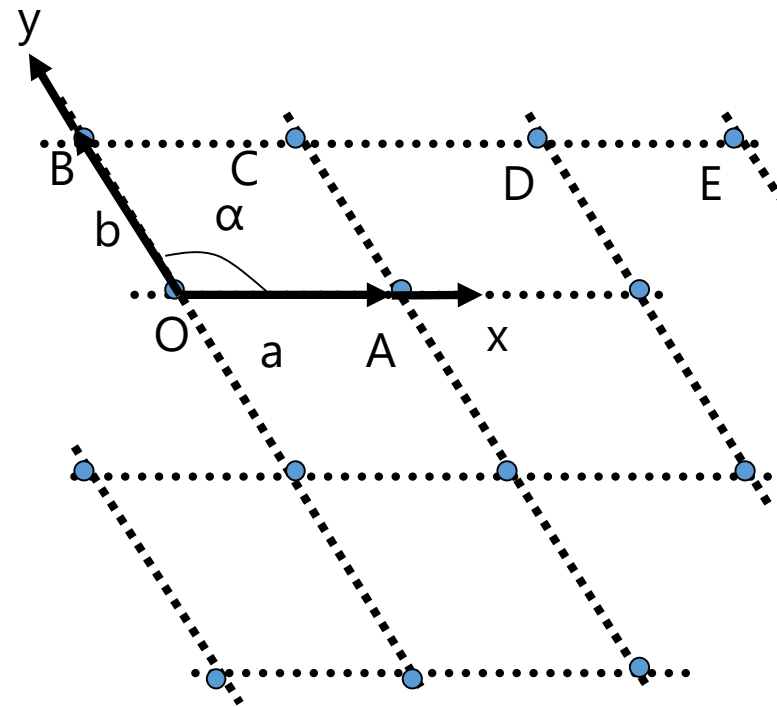


Crystal lattice and
structure of Platinum

Crystal lattice



- An infinite array of points in space,
- Each point has identical surroundings to all others.
- Arrays are arranged in a periodic manner.

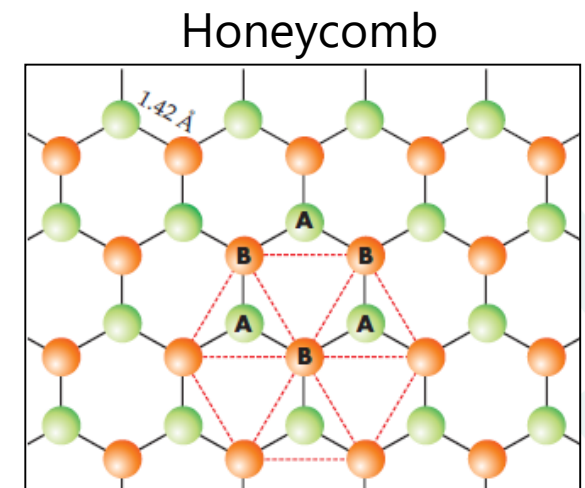
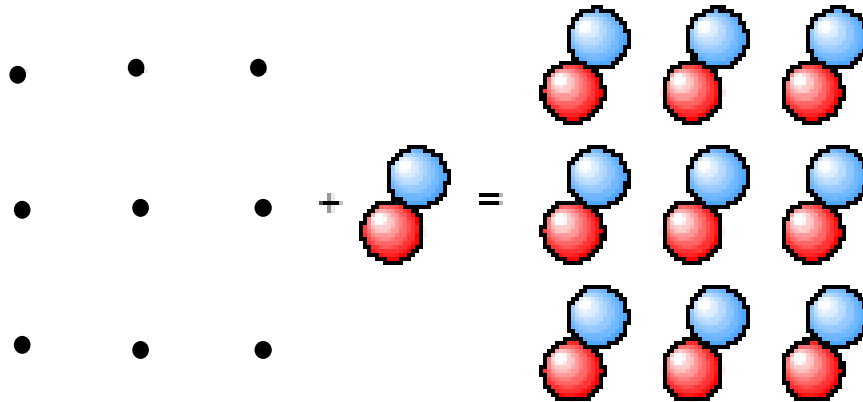


Crystal Structure



- Crystal structures can be obtained by attaching atoms, groups of atoms or molecules which are called basis (motif) to the lattice sites of the lattice point.

Crystal Structure = Crystal Lattice • + Basis 



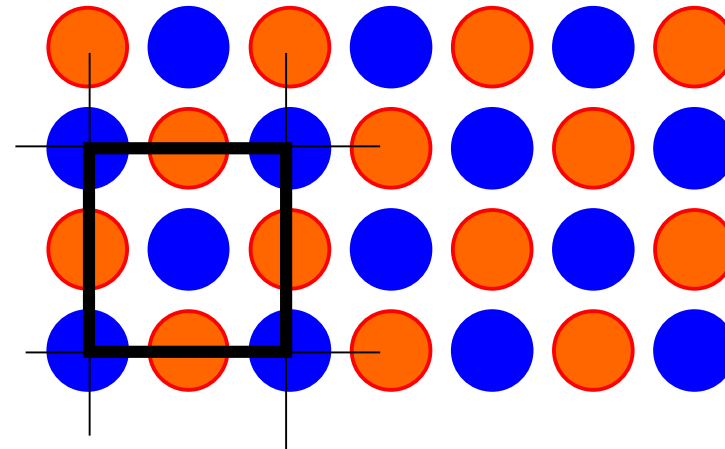
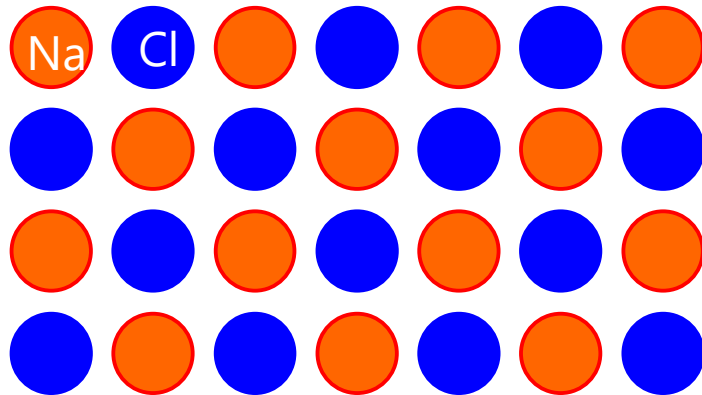
Unit cell



The **fundamental building block** that repeats its own dimensions in various directions and gives crystal structure – **Unit cell**

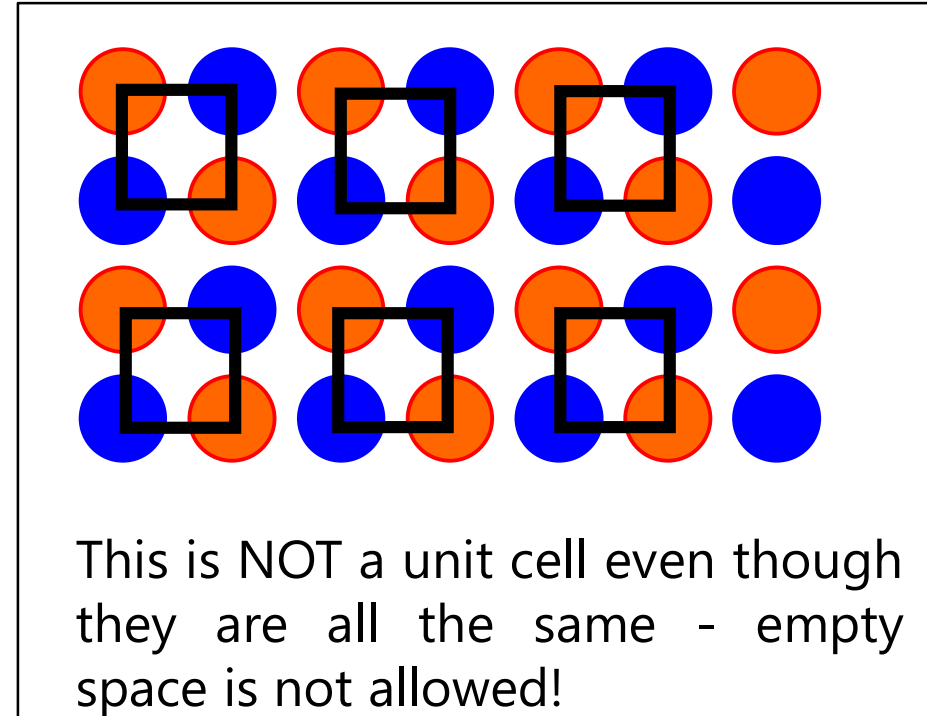
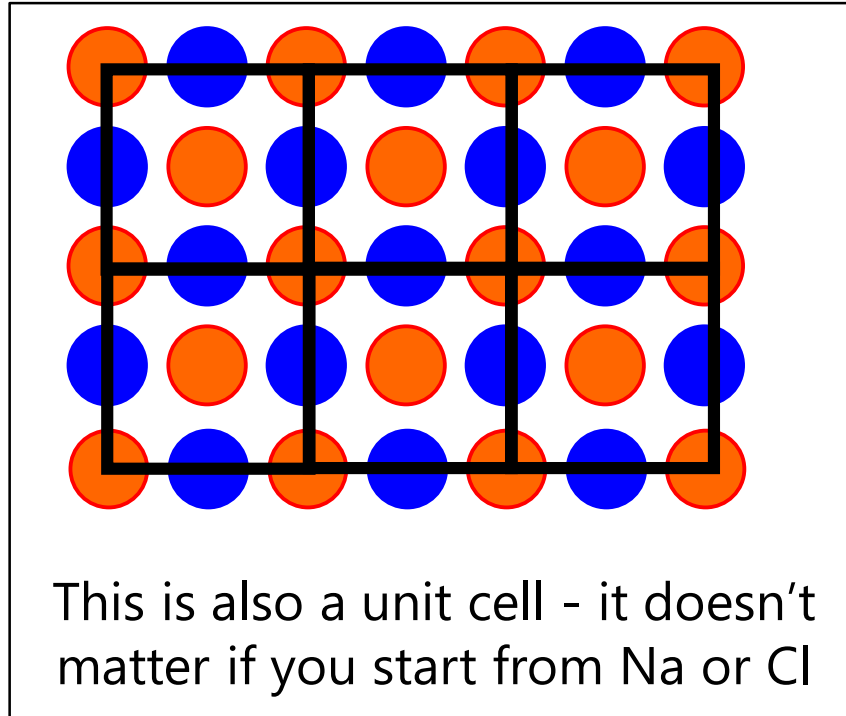
2D example - rocksalt (sodium chloride, NaCl)

points with *identical environments*



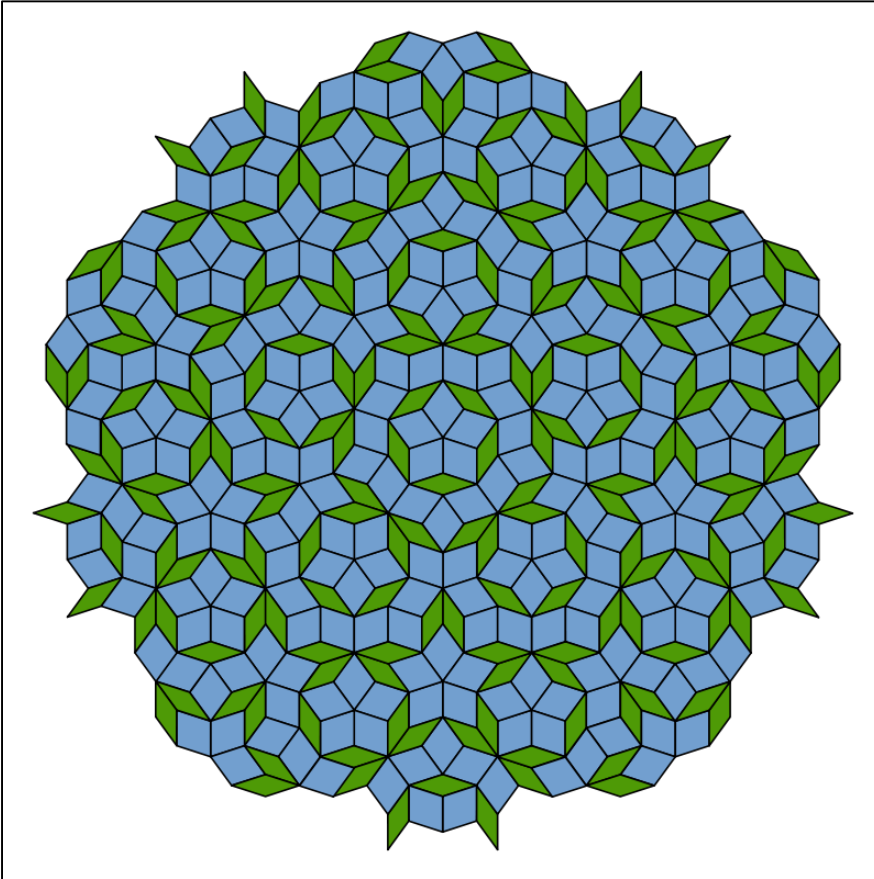
Choice of origin is arbitrary - lattice points need not be atoms - **but** unit cell size should always be the same.

Unit cell

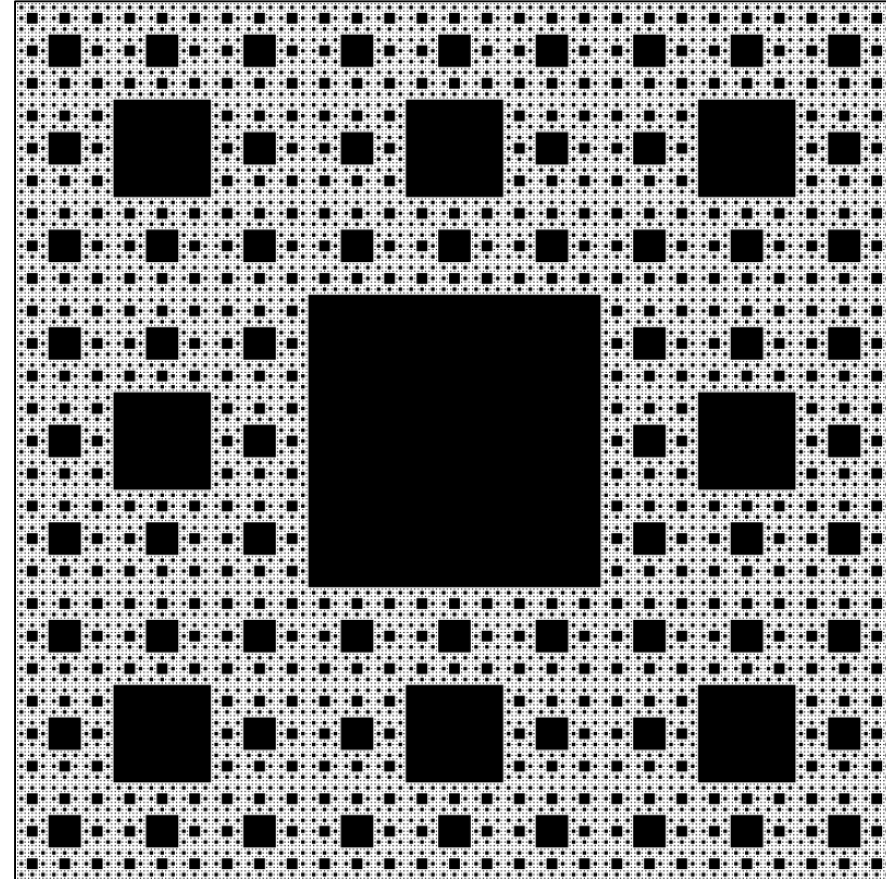


- **Unit cells must link up - cannot have gaps between adjacent cells**
- **All unit cells must be identical**
- **Unit cells must show the full symmetry of the structure**

do you think these structures are crystals?



Quasi Crystal - aperiodic, yet exhibit orientational symmetry (AlCuFe, Al₆CuLi₃)



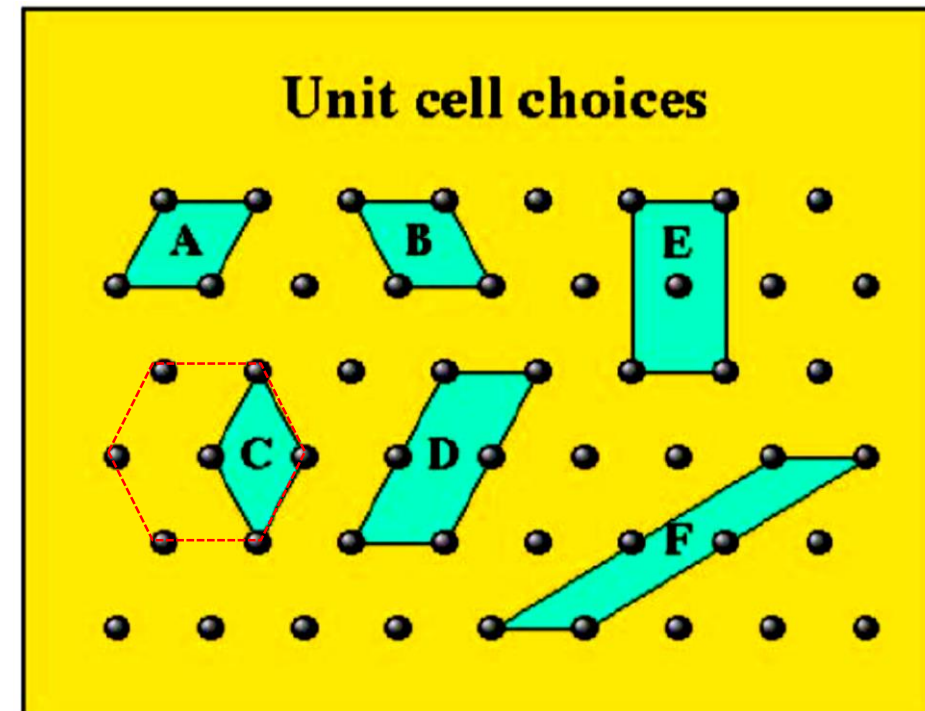
Fractals - self-similarity (e.g. silica aerogel)

Conventional & Primitive Unit cell



➤ The **minimum volume unit cell is known as primitive unit cell**. Primitive unit cell always consists only ONE basis. There are several possible choices of primitive cell for any crystal

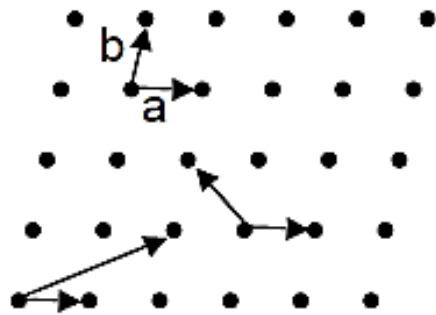
- Which unit cell is good choice?
- A, B, C are primitive unit cell, however D, E, F are not. Why?
- The volume of A, B, C are same but the choice of origin is different. It doesn't matter.



Primitive Unit cell vector



Primitive Vectors



$$\mathbf{R} = u\mathbf{a} + v\mathbf{b} + w\mathbf{c}$$

u, v, w are integers.

$\mathbf{a}, \mathbf{b}, \mathbf{c}$ are called primitive, if the volume of the cell is the smallest possible

Geometric object defined by PV is termed as unit cell

$$V_c = \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$$

Rules:

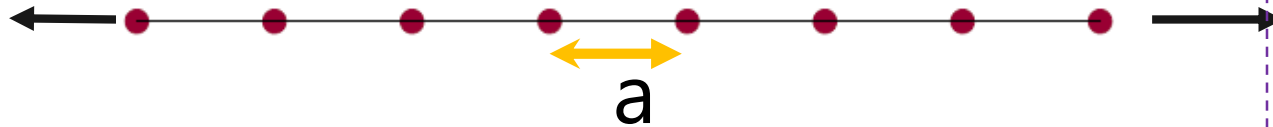
1. $\mathbf{a}$ to be shortest period of the lattice
2. $\mathbf{b}$ to shortest period and non-parallel to $\mathbf{a}$
3. $\mathbf{c}$ to be shortest period & non co-planner with $\mathbf{b}$ and $\mathbf{a}$.

A fundamental concept in the description of crystalline solids is that of a "Bravais lattice".

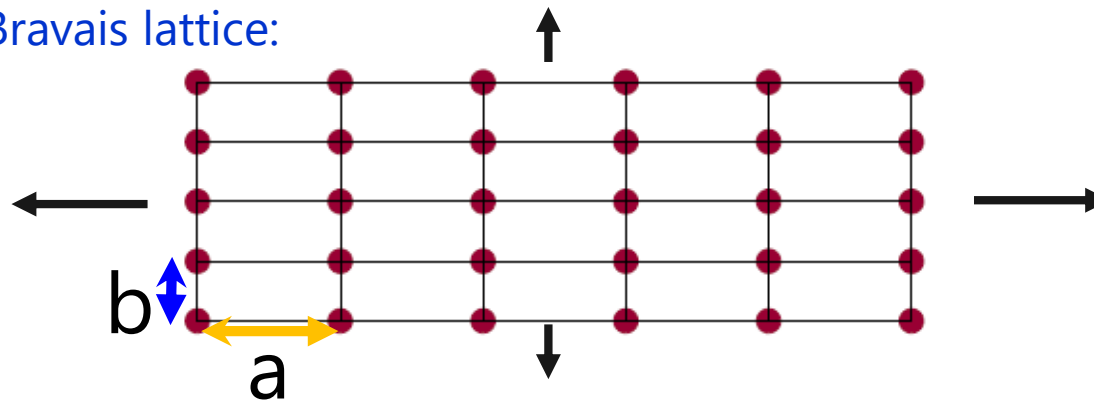
Bravais Lattice



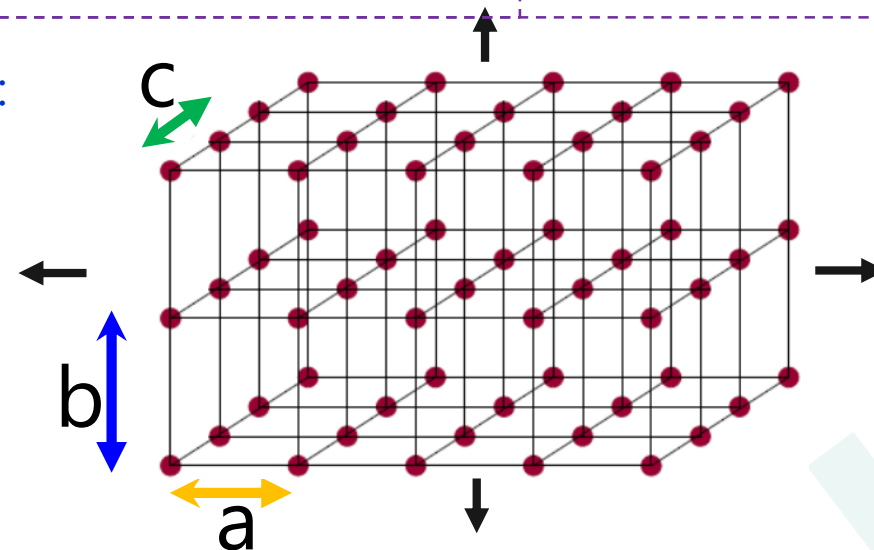
A 1D Bravais lattice:



A 2D Bravais lattice:



A 3D Bravais lattice:

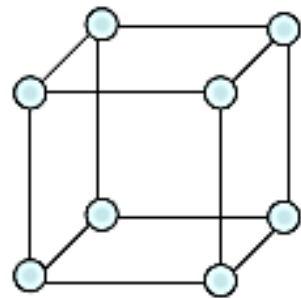


Types of Bravais Lattice

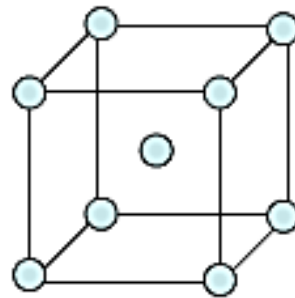


There are just 14 three dimensional crystal lattices (7 main and their subdivisions) with different symmetries - known as Bravais Lattice

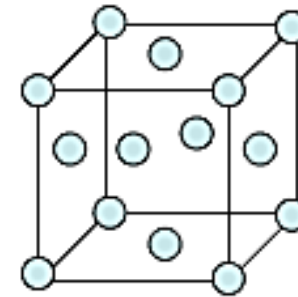
Seven main divisions: Cubic, Tetragonal, Orthorhombic, Hexagonal, Trigonal, Monoclinic, and Triclinic



Simple Cubic



Body Centered



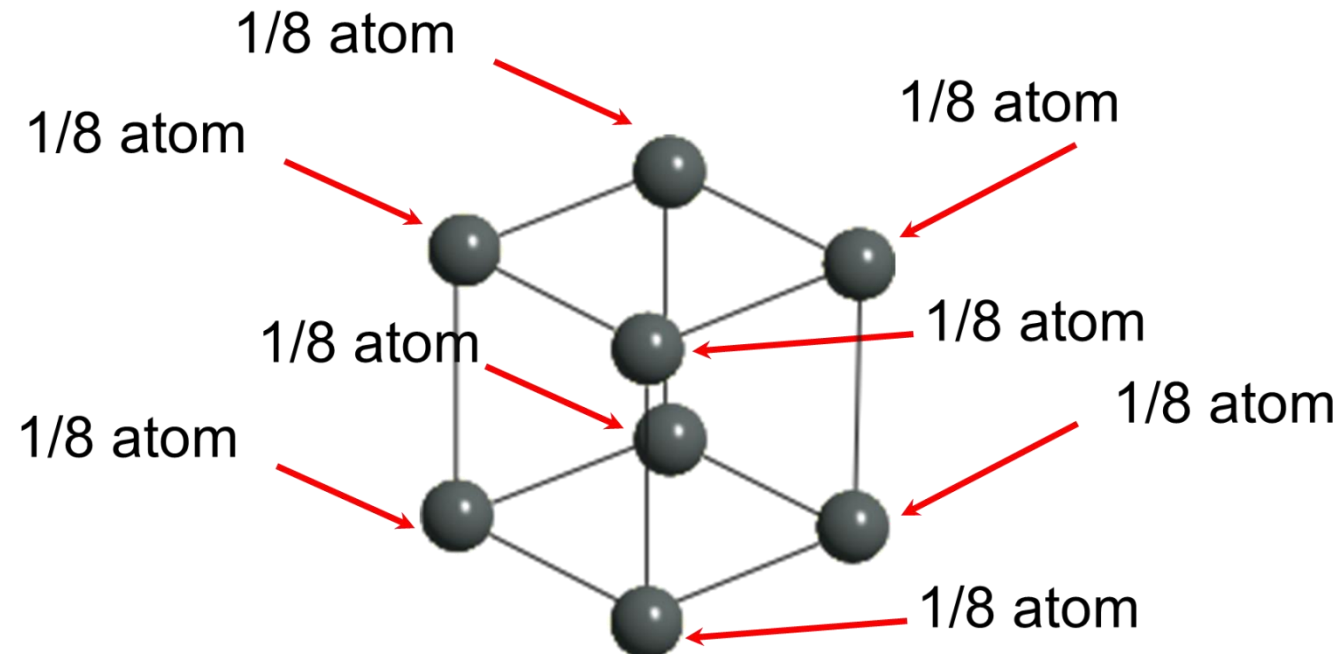
Face Centered

⌘ **Study at home: Miller Indices**

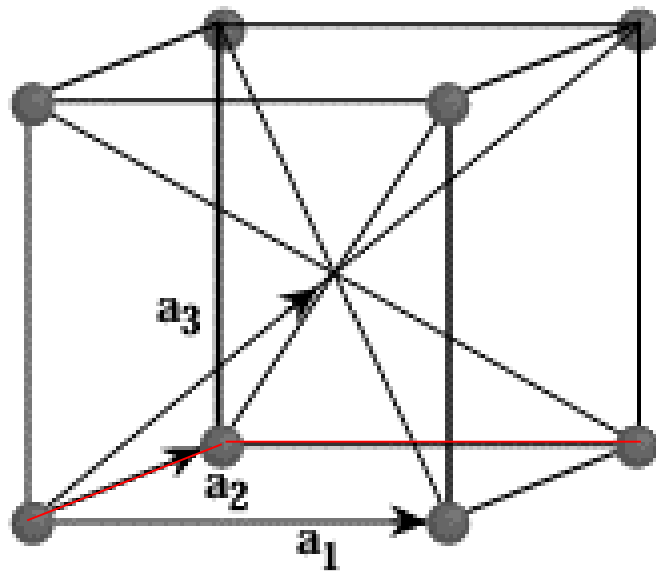
Primitive unit cells of simple cubic



- We can also simply count the atoms we see in one unit cell.
- But we have to keep track of how many unit cells share these atoms.

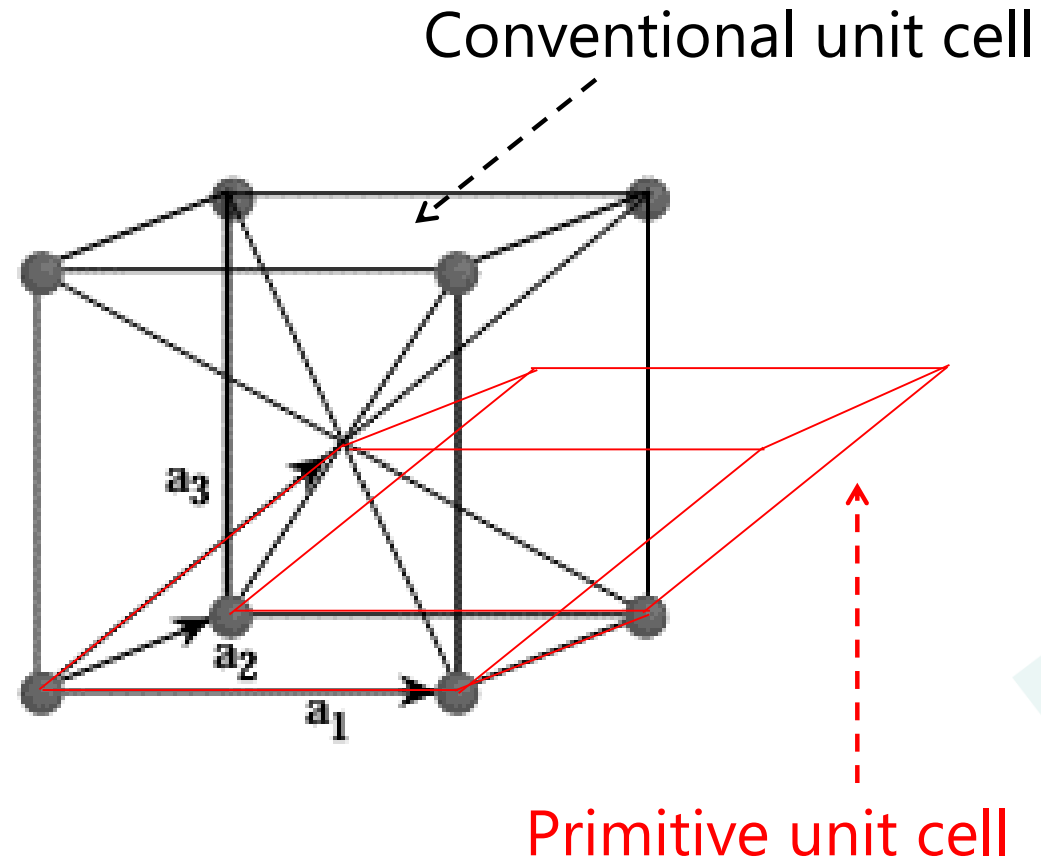


Conventional & primitive unit cell of bcc



BCC

How many atoms in the conventional unit cell?



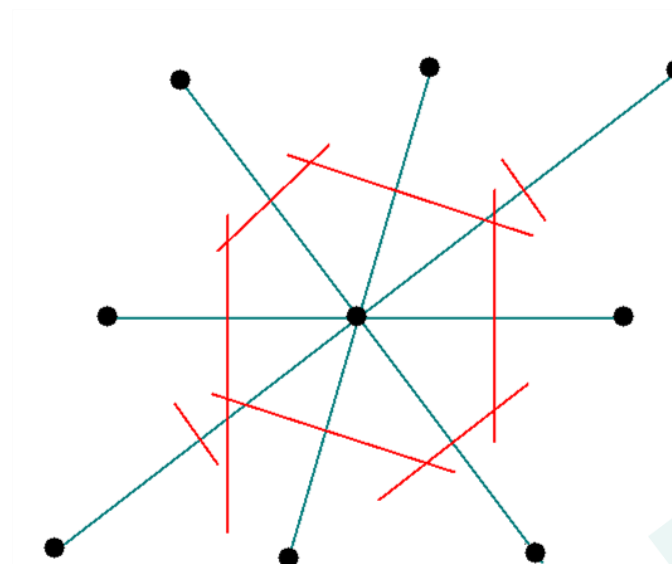
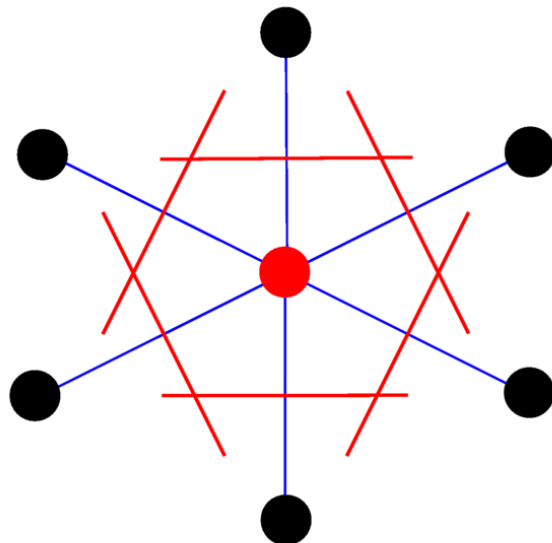
A special primitive cell: Wigner-Seitz cell



Wigner-Seitz cell: smallest possible primitive cell, which consist of one lattice point and all the surrounding space closer to it than to any other point.

The cell may be chosen by first picking a lattice point. Then, lines are drawn to all nearby (closest) lattice points. At the midpoint of each line, another line (or a plane, in 3D) is drawn normal to each of the first set of lines.

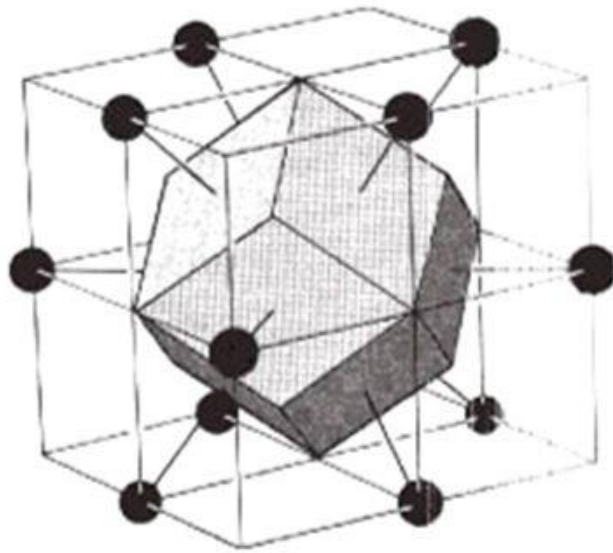
Example: 2D case



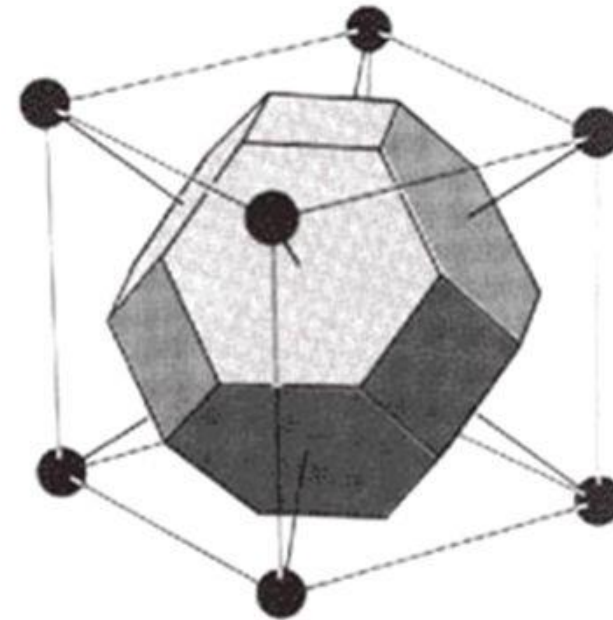
Wigner-Seitz cell in 3D



3 Dimensional Wigner-Seitz Cells



**Face Centered Cubic
Wigner-Seitz Cell**



**Body Centered Cubic
Wigner-Seitz Cell**